

Matematisk Statistik

Ekstra slides

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lm og summary

```
lmfit <- lm(logGlu ~ logColi)
summary(lmfit)
```

eller

```
lmfit <- lm(logGlu ~ 1 + logColi)
summary(lmfit)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.8872	0.1334	-29.14	<2e-16 ***
logColi	0.8494	0.0286	29.69	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3094 on 96 degrees of freedom

Multiple R-squared: 0.9018, Adjusted R-squared: 0.9008

F-statistic: 881.7 on 1 and 96 DF, p-value: < 2.2e-16

Sammenlign:

$$\alpha \leftarrow \hat{\alpha} = -3.887245 \quad \beta \leftarrow \hat{\beta} = 0.8493586 \quad \sigma^2 \leftarrow s_{02}^2 = 0.09575192$$

lm og summary

Hvis output fra `summary` gemmes, kan det bruges i sammenhænge.

```
sumfit <- summary(lmfit)
sumfit$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.8872452	0.13339304	-29.14129	1.802966e-49
logColi	0.8493586	0.02860352	29.69420	3.546582e-50

Koefficienterne kan også findes ved at bruge `coef` på `lmfit`

```
coef(lmfit)
```

(Intercept)	logColi
-3.8872452	0.8493586

Konfidensintervaller for linjens værdi og prædiktionsintervaller

```
nyData = data.frame(logColi = c(4.9,5.7,6.8))  
predict(lmfit,nyData,interval = "confidence")  
predict(lmfit,nyData,interval = "prediction")
```

Output fra ovenstående kode bliver

```
> predict(lmfit,nyData,interval = "confidence")  
      fit      lwr      upr  
1 0.2746120 0.2091721 0.3400519  
2 0.9540989 0.8633519 1.0448458  
3 1.8883933 1.7455389 2.0312478  
> predict(lmfit,nyData,interval = "prediction")  
      fit      lwr      upr  
1 0.2746120 -0.3430939 0.8923178  
2 0.9540989  0.3332018 1.5749959  
3 1.8883933  1.2577702 2.5190165
```

lm og summary

```
rotationer <- c(170,170,170,170,530,530,530,530,880,880,880,880)
masse <- c(3.89,3.95,3.85,3.91,...,3.30,3.26,3.30,3.34)
lmfit_m2 <- lm(masse ~ 1 + rotationer)
summary(lmfit_m2)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.058e+00	2.897e-02	140.08	< 2e-16 ***
rotationer	-8.445e-04	4.819e-05	-17.52	7.78e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04839 on 10 degrees of freedom
Multiple R-squared: 0.9685, Adjusted R-squared: 0.9653
F-statistic: 307.1 on 1 and 10 DF, p-value: 7.78e-09

Estimator: $\hat{\alpha} = 4.05810$ $\hat{\beta} = -0.00084$

Prædikterede værdier: $4.05810 - 0.00084 \cdot 170 = 3.915, 3.6129$ og 3.3189

Invers regression med 1 observation af logGlu på 1.05.

```
lmfit=lm(logGlu~logColi)

invRegLog=inversReg(lmfit,1.05)
invRegLog$Konfidensinterval
exp(invRegLog$Konfidensinterval)
```

```
5.085127 6.552178
161.6004 700.7684
```

Invers regression med 4 observationer af logGlu på 1.05.

```
invRegLog=inversReg(lmfit,rep(1.05,4))
invRegLog$Konfidensinterval
exp(invRegLog$Konfidensinterval)
```

```
5.439088 6.198216
230.2321 491.8709
```

Hypotesetest med **summary** - lineær regression

I modellen M_2 for simpel lineær regression ønsker vi til slut at undersøge

$$H_{03}^* : \alpha = \alpha_0 = \ln 60 = 4.0943.$$

Vi betragter de transformerede variable $\tilde{X}_i = X_i - \ln 60$. I M_2 vil der gælde at

$$\tilde{X}_i \sim N(\tilde{\alpha} + \beta t_i, \sigma^2) \quad i = 1, \dots, n \quad + \text{ uafhængighed}$$

hvor $\tilde{\alpha} = 0$, således at hypotesen kan formuleres som

$$H_{03}^* : \tilde{\alpha} = 0.$$

Hypotesetest med **summary** - lineær regression

```
masse_minus_ln60 <- masse - log(60)
lmfit_m2 <- lm(masse_minus_ln60 ~ 1 + rotationer)
summary(lmfit_m2)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.625e-02	2.897e-02	-1.251	0.239
rotationer	-8.445e-04	4.819e-05	-17.524	7.78e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04839 on 10 degrees of freedom

Multiple R-squared: 0.9685, Adjusted R-squared: 0.9653

F-statistic: 307.1 on 1 and 10 DF, p-value: 7.78e-09

Lineær regression med kendt afskæring

Når vi skal estimere i M_3^* skal skrive 0 i modelformlen, for at specificere at afskæringen er $\alpha_0 = 0$.

```
lmfit_m3star <- lm(masse_minus_ln60 ~ 0 + rotationer)
summary(lmfit_m3star)
```

Parametertabellen bliver

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
rotationer -8.973e-04  2.383e-05  -37.66 5.59e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04962 on 11 degrees of freedom
Multiple R-squared:  0.9923,    Adjusted R-squared:  0.9916
F-statistic: 1418 on 1 and 11 DF,  p-value: 5.588e-13
```

Lineær regression med kendt afskæring

Konfidensintervaller: findes med **confint**

```
confint(lmfit_m2)
confint(lmfit_m3star)
```

```
> confint(lmfit_m2)
              2.5 %          97.5 %
(Intercept) -0.100799208  0.0283015671
rotationer   -0.000951861 -0.0007371102
> confint(lmfit_m3star)
              2.5 %          97.5 %
rotationer -0.0009497517 -0.00084487
```

Bredder af konfidensintervallerne:

```
confint(lmfit_m2)[2,2]-confint(lmfit_m2)[2,1]
confint(lmfit_m3star)[2]-confint(lmfit_m3star)[1]
```

```
> confint(lmfit_m2)[2,2]-confint(lmfit_m2)[2,1]
[1] 0.0002147508
> confint(lmfit_m3star)[2]-confint(lmfit_m3star)[1]
[1] 0.0001048818
```